12:01

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✅
Correct!

Jupyter/Colab notebooks use the .ipynb (Interactive Python Notebook) extension.

Question 3 of 3

Shortcut to run a cell?

- ○ Ctrl + C
- ○ Alt + F4
- ◉ Shift + Enter
- ○ Esc

✅
Correct!

Shift + Enter runs the current cell and moves to the next one.

🎉 Quiz Complete!

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12:02

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upper()

✅
Correct!
max() returns the largest item in an iterable.

Question 3 of 3

Why do we calculate average yield?

- To pick the worst tree
- To summarize performance
- To waste time
- To complexify data

✅
Correct!
Averages give a single number summary of the overall performance.

🎉 Quiz Complete!

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12:02

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plt.chart()

✅
Correct!
plt.bar() creates bar charts.

Question 3 of 3

What must you call to display the plot?

- (●) plt.show()
- ( ) plt.render()
- ( ) plt.view()
- ( ) plt.display()

✅
Correct!
plt.show() renders the figure window.

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12:03

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np.list()

✅
Correct!

np.array() converts a list into a NumPy array.

Question 3 of 3

Can NumPy arrays calculate element-wise?

- (●) Yes
- ( ) No
- ( ) Only sums
- ( ) Only multiplication

✅
Correct!

Yes, operations like array1 + array2 happen element-wise automatically.

🎉 Quiz Complete!

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✅
Correct!

Transposing swaps rows and columns.

Question 3 of 3

Why are matrices important in ML?

- They aren't
- Data is represented as matrices
- For decoration
- Only for math

✅
Correct!

Datasets are matrices; operations like dot products power neural networks.

🎉 **Quiz Complete!**

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12:03

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Adds to last

✅ Correct!

The scalar is broadcasted to match the array shape.

Question 3 of 3

Broadcasting rule:

- (●) Trailing dims must match or be 1
- ( ) Leading dims match
- ( ) No rules
- ( ) Only for 1D

✅ Correct!

Dimensions are compatible if they are equal or one is 1.

🎉 Quiz Complete!

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12:04

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Only average

✅ **Correct!**

Covariance shows if variables increase/decrease together.

Question 3 of 3

Standard deviation measures:

- Average
- Spread of data
- Total sum
- Minimum

✅ **Correct!**

Std dev shows how dispersed data is from the mean.

🎉 **Quiz Complete!**

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✅ **Correct!**

Pandas is the primary library for data manipulation.

Question 3 of 3

How do you view the first 5 rows?

- .tail()
- .view()
- (●) .head()
- .top()

✅ **Correct!**

.head() shows the first 5 rows by default.

🎉 Quiz Complete!

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12:04

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df.append()

✅ Correct!

Direct assignment creates a new column: df['new_col'] = data

Question 3 of 3

Removing a column:

df.remove()

df.drop('col', axis=1)

df.delete()

df.pop()

✅ Correct!

Use drop() with axis=1 for columns.

🎉 Quiz Complete!

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12:05

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File size

✅
Correct!

info() provides a concise summary of the DataFrame.

Question 3 of 3

Best plot for value distribution:

- Line plot
- Histogram
- Scatter
- Pie

✅
Correct!

Histograms visualize the frequency distribution.

🎉 Quiz Complete!

6
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12:05

loc is deprecated

✅ **Correct!**

loc uses labels, iloc uses integer positions.

Question 3 of 3

Query method uses:

- SQL syntax
- String expressions
- Only numbers
- JSON

✅ **Correct!**

df.query('column > 5') uses string expressions.

🎉 Quiz Complete!

6
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12:07

fillna() is used to:

- Delete rows
- Replace missing values
- Find NaN
- Count missing

✅ **Correct!**

fillna() replaces NaN with specified values.

Question 3 of 3

dropna() will:

- Fill missing
- Remove rows/columns with NaN
- Count NaN
- Nothing

✅ **Correct!**

dropna() removes rows or columns containing NaN.

🎉 Quiz Complete!

6
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After grouping, you can:

- (●) Apply aggregation functions
- ( ) Only print
- ( ) Delete data
- ( ) Rename

✅ **Correct!**

Apply sum(), mean(), count(), etc. after groupby().

Question 3 of 3

Pivot tables:

- ( ) Are Excel-only
- (●) Reshape and aggregate data
- ( ) Delete data
- ( ) Same as groupby

✅ **Correct!**

Pivot tables reshape data with aggregations.

🎉 **Quiz Complete!**

6 / 3

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12:08

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Linear Regression

Question 2 of 3

Which function plots a line?

- .bar()
- .scatter()
- .plot()
- .hist()

✅ **Correct!**

.plot() is the standard function for line charts.

Question 3 of 3

Why visualize data?

- To use more colors
- To find patterns and trends
- To hide data
- For fun

✅ **Correct!**

Visualization helps humans spot patterns and outliers quickly.

🎉 **Quiz Complete!**

6 / 3

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12:08

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Box plots reveal:

- (●) Outliers and quartiles
- ( ) Only average
- ( ) Time trends
- ( ) Nothing

✅ **Correct!**

Box plots show median, quartiles, and outliers.

Question 3 of 3

Heatmaps are best for:

- ( ) Single values
- (●) Correlation matrices
- ( ) Line trends
- ( ) Text

✅ **Correct!**

Heatmaps visualize correlation or intensity matrices.

🎉 **Quiz Complete!**

6 / 3

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12:08

Visualizatic

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Cross-

sns.heatmap() is good for:

- Time series
- Correlation matrices
- Bar charts
- Pie charts

✅ **Correct!**

Heatmaps visualize matrix values well.

Question 3 of 3

Seaborn works well with:

- Lists
- Pandas DataFrames
- Dictionaries
- Sets

✅ **Correct!**

It integrates directly with DataFrames.

🎉 **Quiz Complete!**

6 / 3

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Data preparation includes:

- Only loading data
- Cleaning, transforming, splitting
- Just visualization
- Only testing

✅ **Correct!**

Preparation = cleaning, feature engineering, splitting.

Question 3 of 3

Why evaluate models?

- It's optional
- To measure performance
- To waste time
- For fun

✅ **Correct!**

Evaluation tells us if the model works well.

🎉 **Quiz Complete!**

6 / 3

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12:09

Unsupervised learning example:

- Regression
- Clustering (K-Means)
- Classification
- Decision Tree

✅ Correct!

Clustering groups unlabeled data.

Question 3 of 3

Predicting house price is:

- Supervised Regression
- Unsupervised
- Classification
- Clustering

✅ Correct!

Price is a continuous label.

🎉 Quiz Complete!

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Types

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- Too simple model
- Memorizing training data
- Perfect model
- No learning

✅ Correct!

Overfitting = high train accuracy, low test accuracy.

Question 2 of 3

Underfitting means:

- Perfect fit
- Too complex
- Too simple, can't learn patterns
- Good generalization

✅ Correct!

Underfitting = model too simple to capture patterns.

Question 3 of 3

Bias-Variance tradeoff:

- Not important
- Balance between underfitting and overfitting
- Only for deep learning
- Deprecated

✅ Correct!

Balance bias (underfitting) and variance (overfitting).

🎉 **Quiz Complete!**

6 / 3

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12:09

REVA UNIVERSITY | AI/ML in Agriculture | Instructor

Question 1 of 3

K-Fold CV splits data into:

- 2 parts
- K equal parts
- Random parts
- No split

✅ Correct!

K-Fold divides data into K folds, trains K times.

Question 2 of 3

Advantage of CV:

- Faster
- More robust performance estimate
- Uses less data
- Simpler

✅ Correct!

CV gives better estimate by averaging K runs.

Question 3 of 3

Stratified CV ensures:

- Random split
- Class proportions maintained
- No split
- Faster training

✅ Correct!

Stratified keeps class distribution in each fold.

🎉 Quiz Complete!

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Crop Yield Prediction Models
Regression Techniques for Yield Forecasting
Weather Data Integration
Soil Data Integration
Crop Health Assessment
Plant Disease

By the model

Before training

After training

Randomly

✅ **Correct!**

They configure the training process.

Question 2 of 3

Grid Search is used for:

Data cleaning

Finding best hyperparameters

Visualization

Deployment

✅ **Correct!**

It exhaustively searches hyperparameter combinations.

Question 3 of 3

Learning rate is a:

Parameter

Hyperparameter

Metric

Feature

✅ **Correct!**

It controls step size during optimization.

🎉 **Quiz Complete!**

6 / 3

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Weather Data Integration

Soil Data Integration

Crop Health Assessment

Plant Disease

Import sklearn.linear

None

✅ Correct!

Classes are organized in submodules.

Question 2 of 3

fit() method does what?

- Predicts
- Trains the model
- Scores
- Splits

✅ Correct!

fit() trains the algorithm on data.

Question 3 of 3

predict() method:

- Trains
- Generates predictions on new data
- Evaluates
- Cleans

✅ Correct!

predict() uses the trained model.

🎉 **Quiz Complete!**

6 / 3

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12:11

REVA UNIVERSITY

AI/ML in Agriculture

Instructor

Question 1 of 3

Linear regression predicts which type of variable?

- Categorical
- Continuous/Numeric
- Images
- Text

✅ Correct!

Regression predicts continuous values like price, height, or yield.

Question 2 of 3

What represents the relationship in simple linear regression?

- A curve
- A straight line
- A circle
- A dot

✅ Correct!

It fits a straight line (y = mx + c) to the data.

Question 3 of 3

What is the 'best fit' line?

- Longest line
- Line minimizing error
- Highest line
- Thickest line

✅ Correct!

It minimizes the distance (error) between data points and the line.

🎉 Quiz Complete!

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Question 1 of 3

RMSE stands for:

- Root Mean Square Error
- Random Mean Score
- Rate My Squared Error
- None

✅ Correct!

RMSE = Root Mean Squared Error, measures prediction error.

Question 2 of 3

R^2 Score ranges:

- 0 to 100
- 0 to 1
- -∞ to 1
- Always positive

✅ Correct!

R^2 can be negative for poor models, 1 is perfect.

Question 3 of 3

Lower MAE means:

- Worse model
- Better model
- No difference
- Invalid

✅ Correct!

MAE = Mean Absolute Error; lower is better.

🎉 Quiz Complete!

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12:12

Question 2 of 3

Function used in Logistic Regression:

- Linear
- Sigmoid
- ReLU
- Step

✅ **Correct!**

Sigmoid maps output to 0-1 probability.

Question 3 of 3

Output of Logistic Regression:

- Any number
- Probability (0 to 1)
- Integer
- String

✅ **Correct!**

It outputs probability of class membership.

🎉 Quiz Complete!

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12:13

Question 2 of 3

Pruning is:

- Growing tree
- Trimming tree to avoid overfitting
- Deleting data
- Training

✅ **Correct!**

Pruning removes branches to prevent overfitting.

Question 3 of 3

Leaf nodes contain:

- More splits
- Final predictions
- Nothing
- Only errors

✅ **Correct!**

Leaf nodes are terminal and make predictions.

🎉 Quiz Complete!

6 / 3

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12:13

Question 2 of 3

Can we use it for feature selection?

- (●) Yes
- ( ) No
- ( ) Maybe
- ( ) Only for images

✅ **Correct!**

Yes, drop low importance features.

Question 3 of 3

Tree models provide importance naturally?

- (●) Yes
- ( ) No
- ( ) Never
- ( ) Only SVM

✅ **Correct!**

Trees calculate split importance automatically.

🎉 Quiz Complete!

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